QUANTITATIVE ANALYSIS • LABOR POLICY • 2026 PORTFOLIO REVISION

Recreational Marijuana Legalization and Manufacturing Work Hours

State-level econometric analysis | Originally submitted for Econometrics, December 2024

An exploratory test of whether recreational marijuana legalization is associated with average weekly hours worked by manufacturing production employees across U.S. states.

QUESTION AND APPROACH

Using 2021 cross-sectional data for all 50 states, the analysis estimates ordinary least squares models with controls for education, obesity, minimum wage, family size, region, and demographic composition. It includes a logged specification and diagnostic tests for heteroskedasticity and multicollinearity.

CONTRIBUTION

In the main specification, legalization status is associated with approximately 1.2 fewer average weekly manufacturing payroll hours, holding selected controls constant. The estimate is statistically significant at the 10 percent level. A logged model yields a similar directional association. The analysis treats both results as exploratory—not causal.

SELECTED TAKEAWAYS

- A single-year state comparison cannot establish that legalization changed work hours; unmeasured state differences may drive the result.
- The outcome covers manufacturing production employees, not productivity, total employment, or work hours for all workers.
- The project demonstrates the value of stating what a model cannot answer and identifying stronger next-step designs using longitudinal or individual-level data.

SKILLS DEMONSTRATED

- Stata, data assembly, model specification, robustness checks, and interpretation of regression output
- Research synthesis, limitation analysis, and communication of uncertain evidence for policy audiences

Portfolio note: Academic research revised for a professional portfolio. Full paper available upon request.